

Question	Marking Guidance	Mark	Comments
5.1	$\text{ATP} \longrightarrow \text{ADP} + \text{P}_i + \text{H}_2\text{O}$ ☐ $\text{ATP} + \text{H}_2\text{O} \longrightarrow \text{ADP} + \text{P}_i$ ☐ $\text{ADP} + \text{P}_i \longrightarrow \text{ATP} + \text{H}_2\text{O}$ ☒ $\text{ADP} + \text{P}_i + \text{H}_2\text{O} \longrightarrow \text{ATP}$ ☐	1	
5.2	Human ATP synthase has a different <u>tertiary</u> structure to bacterial ATP synthase OR Human ATP synthase has a different <u>shape</u> <u>active site</u> to bacterial ATP synthase OR Antibiotic cannot enter human cells/mitochondria OR Antibiotic not <u>complementary</u> (to human ATP synthase);	1	
5.3	0.75;; One mark for showing $30 \text{ g} = 0.03 \text{ kg}$; One mark for showing 0.025 mg g^{-1}	2	
5.4	Answer in range 97.0 – 97.8%;; OR Answer in range 3288 – 4368%;;	2	1 mark for correct $\log_{10}$ readings from graph converted to actual numbers (16.98 – 19.50 and 660.7 – 758.6)

5.5	1. (From Fig 2) New/old antibiotic does not kill all bacteria; OR (From Fig 2) Some bacteria are resistant to the new/old antibiotic; 2. Resistant bacteria will reproduce to produce (more) resistant bacteria; 3. (Use of both) one antibiotic will kill bacteria resistant to the other antibiotic; OR Unlikely that bacteria are resistant to both the new and the old antibiotic; OR Use of both antibiotics (likely to) kill all/most bacteria;	3	Accept use of 'A' for 'new antibiotic' and 'R' for 'old antibiotic'. 1. Must relate to the bacteria that are still present – 'some bacteria are killed' or 'the bacteria number is reduced' is insufficient. 2. Accept 'resistant bacteria reproduce to pass on resistance gene/allele' 3. 'Use of both antibiotics will be more effective' is insufficient.
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